

HCLTech Financial RAG Intelligence

Enterprise Retrieval-Augmented Generation (RAG) System for FY26 Financial Filings

GitHub Repository: https://github.com/Prajeeth-12/Finance_RAG_ChatBot

Tech Stack: NVIDIA NIM API · Llama 3 · ChromaDB · Streamlit · Python 3.10

1. Executive Summary & Problem Statement

Dense corporate financial reports (such as quarterly earnings releases and 80+ page filings) present a major bottleneck for financial analysts. Locating specific quarterly metrics (e.g., Revenue, EBIT, or Net Income) requires tedious manual search and is prone to error.

HCLTech Financial RAG Intelligence is an enterprise-grade AI chatbot system designed to parse, index, and query HCLTech’s FY26 quarterly reports (Q1–Q4). Built using **NVIDIA NIM Microservices** and **ChromaDB**, it delivers grounded, instant answers with exact source document citations while enforcing a **0% Math Hallucination Guardrail**.

2. Technical Architecture & Tech Stack

Component	Technology / Model Choice
LLM Generator	NVIDIA NIM API : meta/llama-3.1-8b-instruct (1.5s latency) / meta/llama-3.3-70b-instruct
Embeddings	NVIDIA NIM API : nvidia/nv-embedqa-e5-v5 (1024-dimensional query & passage vectors)
Vector Database	ChromaDB (Persistent vector database storing 244 document chunks)
Text Processing	pypdf text extractor & LangChain RecursiveCharacterTextSplitter (1000 char size, 200 overlap)
Frontend UI	Streamlit (Dark financial dashboard with real-time process execution timeline)

3. Key Engineering Innovations & Guardrails

- **0% Math Hallucination Guardrail:** Enforces strict prompt rules that prevent the LLM from inventing figures, performing arbitrary arithmetic, or dividing annual totals to estimate quarterly metrics.
- **Hybrid Domain Re-Ranking:** Combines vector cosine similarity with financial domain-specific heuristic scoring (quarter scoping + metric keyword boosting + annual penalty weights). This prevents vector search from confusing annual figures with quarterly numbers.
- **NVIDIA NIM Batched Vector Indexing:** Utilizes high-throughput batched API requests (32 chunks per batch) to index 244 financial chunks into ChromaDB in under 5 seconds.
- **Interactive Process Pipeline UI:** Features a live single-line status bar tracking scope parsing, vector search latency, and LLM synthesis timing.

4. End-to-End System Pipeline

1. **PDF Extraction & Chunking:** Extracts text from HCLTech_Q1_FY26.pdf through Q4_FY26.pdf and creates 244 structured text chunks tagged by source quarter.

2. **Vector Store Generation:** Embeds chunks using `nvidia/nv-embedqa-e5-v5` (`input_type="passage"`) and stores them persistently in ChromaDB.
3. **Query Scope Detection:** Parses user question syntax to identify target quarters (Q1–Q4) and target metric types (Revenue, EBIT, Net Income).
4. **Hybrid Candidate Retrieval:** Queries 1024-dim question embeddings (`input_type="query"`), scores top candidates, and extracts top 5 passage chunks.
5. **Grounded LLM Synthesis:** Synthesizes executive financial answers using NVIDIA NIM LLM with zero-temperature strict prompt rules.

5. Verified Financial Benchmarks & Results

User Question	Grounded LLM Output	Primary Source File	Latency
What was HCLTech's revenue in Q4 FY26?	INR Revenue of INR 33,981 Crores , up 0.3% QoQ & up 12.3% YoY.	HCLTech_Q4_FY26.txt (Chunk 6)	1.5s
What was the Net Income in Q1 FY26?	Net Income (NI) at INR 3,843 Crores .	HCLTech_Q1_FY26.txt (Chunk 4)	1.5s
What was EBIT in Q2 FY26?	INR EBIT at INR 5,550 Crores , up 12.3% QoQ & up 3.5% YoY.	HCLTech_Q2_FY26.txt (Chunk 4)	1.5s

6. Repository Links & Setup

Project Quick Start

1. **Clone Repo:** `git clone https://github.com/Prajeeth-12/Finance_RAG_ChatBot.git`
2. **Environment Config:** Set `NVIDIA_API_KEY` in `.env`
3. **Install Dependencies:** `pip install -r requirements.txt`
4. **Launch App:** `streamlit run app.py`

Grounded Financial RAG Assistant · [GitHub: Prajeeth-12/Finance_RAG_ChatBot](https://github.com/Prajeeth-12/Finance_RAG_ChatBot)